

A Lightweight ConvLSTM Architecture for Abnormal Human Activity Recognition in Surveillance Videos

By Sanele Hlabisa & Prof. Jules-Raymond Tapamo

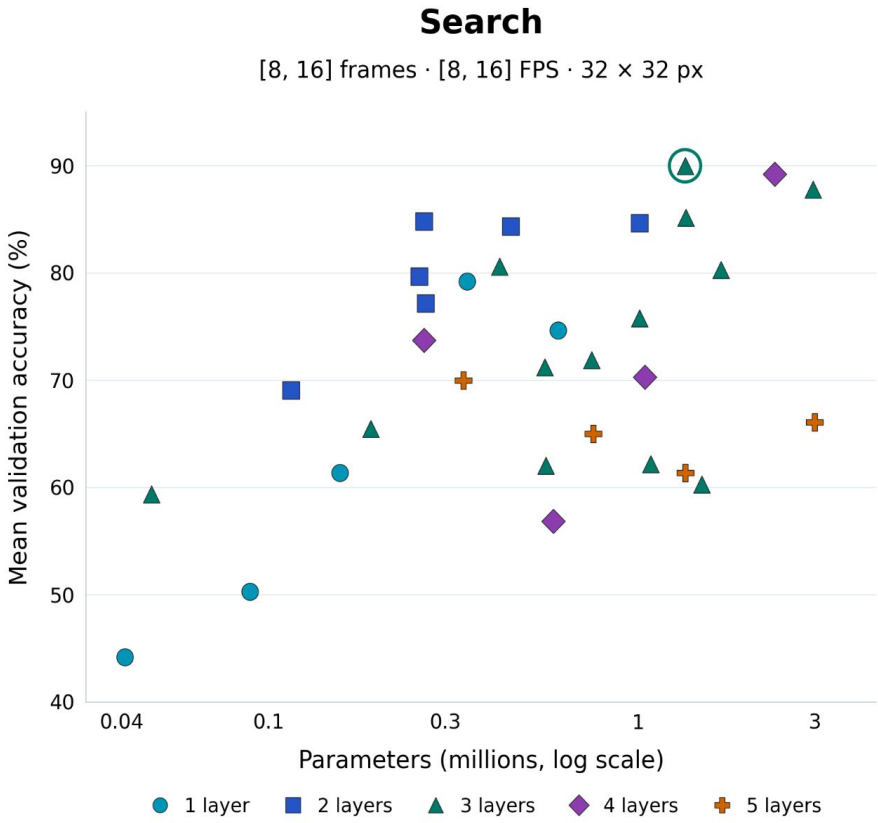
Why This Problem Matters

- **Problem Statement**
  - Cameras record more than people can monitor
  - Fatigue and divided attention can lead to missed events
  - This creates a need for automated activity recognition
- **Existing approaches**
  - 3D CNN: Learns appearance and motion, but is often large
  - ViT: Captures long-range patterns, but is compute-intensive
  - ConvLSTM: Combines spatial features with temporal memory
- **Research question**
  - *Can a compact ConvLSTM maintain strong recognition performance with fewer parameters?*



Proposed Solution and Architecture Search

- **Configurable Architecture**
  - Flat: ConvLSTM[8,8,8]
  - Expanding: ConvLSTM[4,8,16]
  - Contracting: ConvLSTM[16,8,4]
- **Abnormal Activities Dataset**
  - 1,069 video clips
  - 11 activity classes
- **Controlled Search**
  - 32 architectures
  - 4 temporal settings
  - 128 search runs

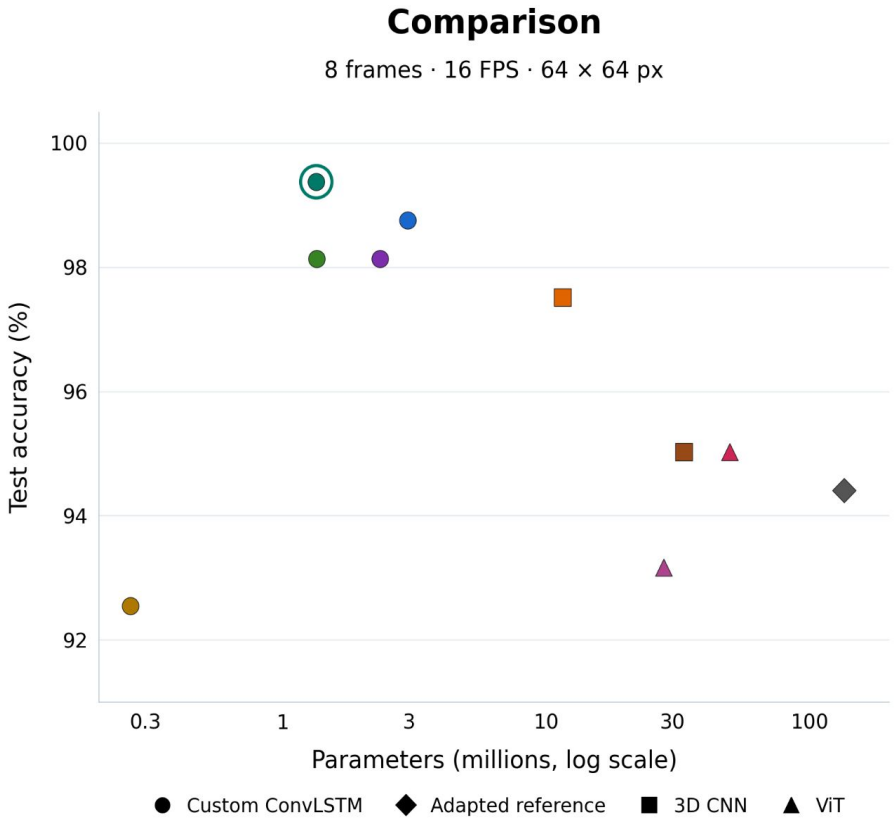


Top 10 search models

- **01 [64,64,128]** 90.00% · 1.34M parameters
- 02 [96,96,96,96] 89.22% · 2.34M parameters
- 03 [128,128,128] 87.81% · 2.97M parameters
- 04 [128,64,64] 85.16% · 1.34M parameters
- 05 [32,64] 84.84% · 0.26M parameters
- 06 [96,96] 84.69% · 1.01M parameters
- 07 [64,64] 84.38% · 0.45M parameters
- 08 [48,48,48] 80.63% · 0.42M parameters
- 09 [96,96,96] 80.31% · 1.67M parameters
- 10 [48,48] 79.69% · 0.25M parameters

Comparison and Future Work

- **Final comparison**
  - Six ConvLSTM models
  - Two 3D CNNs
  - Two ViTs
- **Best Result**
  - 99.38% accuracy, 1.34M params
- **Deployment**
  - Gradio application
- **Future work**
  - Multi-label recognition
  - Continuous overlapping windows



Comparison models

- **01 ConvLSTM[64,64,128]** 99.38% · 1.34M parameters
- 02 ConvLSTM[128,128,128] 98.76% · 2.97M parameters
- 03 ConvLSTM[128,64,64] 98.14% · 1.34M parameters
- 04 ConvLSTM[96,96,96,96] 98.14% · 2.34M parameters
- 05 MC3-18 97.52% · 11.50M parameters
- 06 R3D-18 95.03% · 33.17M parameters
- 07 Swin3D-S 95.03% · 49.52M parameters
- 08 Adapted reference 94.41% · 134.42M parameters
- 09 Swin3D-T 93.17% · 27.86M parameters
- 10 ConvLSTM[32,64] 92.55% · 0.26M parameters